



Film Rental Store (1)

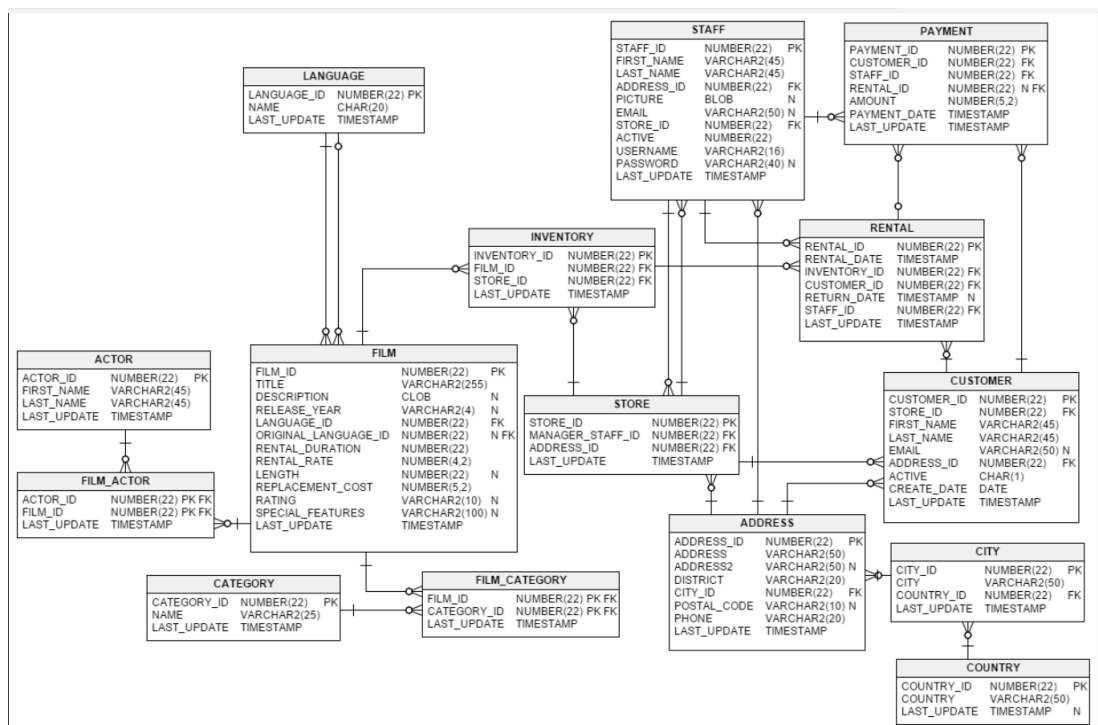
Introduction



This project aims to create a comprehensive and dynamic Film Rental Store system, providing an immersive environment for users to interact with the film rental business domain. The project involves developing a robust platform that efficiently manages the rental process, ensuring a seamless experience for both customers and the business.

Database Schema

▼ ER Diagram



▼ Sakila Database Files

sakila-schema.sql

sakila-data.sql

Schema Location

The Sakila Database (joog.org).

Design Principles

1. All Rest API request and response format must be in JSON.
2. 200, 201, 400, 404, 401, 500, 505 – use appropriate http status code in your response object.
3. Handle the Rest exceptions globally using Rest exception handler for all endpoints.
4. Sample Error Responses are shown in below table, participants to identify error paths for all the Endpoints.
5. Test your Rest Api using Junit and Mockito.
6. Participants should not alter database schema under any circumstances except wherever changes are mentioned explicitly.
7. All GET and POST requests for customer are secured.
8. Alter Picture column of Staff to URL
9. Remove username & password columns of Staff table instead use separate table for security.

Rest API Design

All REST API requests and responses follow the JSON format. The project employs standard HTTP status codes (200, 201, 400, 404, 401, 500, 505) appropriately in response objects to convey the outcome of API requests.

Status Codes

HTTP response codes are used to indicate general classes of success and error.

HTTP Status Quote	Description
200	Successfully processed request.
201	Created
202	Accepted
400	Bad Request
401	Unauthorized
403	Forbidden
404	Not Found
500	Internal Server Error

Endpoints

Endpoints

 Path URL	 HTTP Methods	 Description	 Success Response
<u>/api/films/post</u> —	POST	<u>Add new Film object in DB</u>	String: Record Created Successfully
<u>/api/films/title/{title}</u> —	GET	<u>Search Films by Title</u>	Collection of Films
<u>/api/films/year/{year}</u> —	GET	<u>Search Films by Release Year</u>	Collection of Films
<u>/api/films/duration/gt/{rd}</u> —	GET	<u>Search Films where Rental Duration is greater than {rd}</u>	Collection of Films
<u>/api/films/rate/gt/{rate}</u> —	GET	<u>Search Films where Rental Rate is greater than {rate}</u>	Collection of Films

📍 Path URL	🔗 HTTP Methods	📝 Description	✓ Success Response
<u>/api/films/length/gt/{length}</u> . —	GET	<u>Search Films where Length is greater than {length}.</u>	Collection of Films
<u>/api/films/duration/lt/{rd}</u> . —	GET	<u>Search Films where Rental Duration is lower than {rd}.</u>	Collection of Films
<u>/api/films/rentalrate/lt/{rate}</u> . —	GET	<u>Search Films where Rental Rate is lower than {rate}.</u>	Collection of Films
<u>/api/films/length/lt/{length}</u> . —	GET	<u>Search Films where Length is lower than {length}.</u>	Collection of Films
<u>/api/films/betweenyear/{from}/{to}</u> . —	GET	<u>Search Films which are Released between {from} and {to}.</u>	Collection of Films
<u>/api/films/rating/lt/{rating}</u> . —	GET	<u>Search Films where Rating is lower than {rating}.</u>	Collection of Films
<u>/api/films/rating/gt/{rating}</u> . —	GET	<u>Search Films where Rating is greater than {rating}.</u>	Collection of Films
<u>/api/films/language/{lang}</u> . —	GET	<u>Search Films where Language is {lang}.</u>	Collection of Films
<u>/api/films/countbyyear</u> —	GET	<u>Display number of Films released by each Year</u>	Year & Count of Films
<u>/api/films/{id}/actors</u> —	GET	<u>Find all Actors of a Film by Film id</u>	Collection of Actors
<u>/api/films/category/{category}</u> . —	GET	<u>Find all Films of specified {category}.</u>	Collection of Films
<u>/api/films/{id}/actor</u> —	PUT	<u>Assign Actor to a Film</u>	Collection of Actors

 Path URL	 HTTP Methods	 Description	 Success Response
<u>/api/films/update/title/{id}</u> —	PUT	<u>Update Title of a Film</u>	Film
<u>/api/films/update/releaseyear/{id}</u>	PUT	<u>Update Release Year of a Film</u>	Film
<u>/api/films/update/rentalduration/{id}</u>	PUT	<u>Update Rental Duration of a Film</u>	Film
<u>/api/films/update/rentalrate/{id}</u>	PUT	<u>Update Rental Rate of a Film</u>	Film
<u>/api/films/update/rating/{id}</u> —	PUT	<u>Update Rating of a Film</u>	Film
<u>/api/films/update/language/{id}</u>	PUT	<u>Update Language of a Film</u>	Film
<u>/api/films/update/category/{id}</u>	PUT	<u>Update Category of a Film</u>	Film
<u>/api/actors/post</u> —	POST	<u>Add new Actor object in DB</u>	String: Record Created Successfully
<u>/api/actors/lastname/{ln}</u> —	GET	<u>Search Actors by Last Name</u>	Collection of Actors
<u>/api/actors/firstname/{fn}</u> —	GET	<u>Search Actors by First Name</u>	Collection of Actors
<u>/api/actors/update/lastname/{id}</u> —	PUT	<u>Update Last Name of an Actor</u>	Actor
<u>/api/actors/update/firstname/{id}</u>	PUT	<u>Update First Name of an Actor</u>	Actor

 Path URL	 HTTP Methods	 Description	 Success Response
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<u>/api/actors/{id}/films</u> –	GET	<u>Search Films of an Actor by Actor Id</u>	Collection of Films
– <u>/api/actors/{id}/film</u>	PUT	<u>Assign Film to an Actor</u>	Collection of Films
<u>/api/actors/top10byfilmcount</u> –	GET	<u>Find top 10 Actors by Film Count</u>	Id & Name of Actors with Number of Films
<u>/api/customers/post</u> –	POST	<u>Add new Customer Object in DB</u>	String: Record Created Successfully
<u>/api/customers/lastname/{ln}</u> –	GET	<u>Search Customers by last name</u>	Collection of Customers
<u>/api/customers/firstname/{fn}</u>	GET	<u>Search Customers by first name</u>	Collection of Customers
<u>/api/customers/email/{email}</u> –	GET	<u>Search Customers by email</u>	Customer
<u>/api/customers/{id}/{addressId}</u>	PUT	<u>Assign Address to a Customer</u>	Customer & Address details
<u>/api/customers/city/{city}</u>	GET	<u>Search Customers by City</u>	Collection of Customers with Address
<u>/api/customers/country/{country}</u>	GET	<u>Search Customers by Country</u>	Collection of Customers with Address
<u>/api/customers/active</u>	GET	<u>Search all active Customers</u>	Collection of Customers

 Path URL	 HTTP Methods	 Description	 Success Response
<u>/api/customers/inactive</u>	GET	<u>Search all inactive Customers</u>	Collection of Customers
<u>/api/customers/phone/{phone}</u>	GET	<u>Search Customers by phone number</u>	Customer with Address
<u>/api/customers/update/{id}/{fn}</u>	PUT	<u>Update first name of Customer.</u>	Customer
<u>/api/customers/update/{id}/ln</u>	PUT	<u>Update last name of Customer</u>	Customer
<u>/api/customers/update/{id}/email</u>	PUT	<u>Update email of Customer</u>	Customer
<u>/api/customers/update/{id}/store</u>	PUT	<u>Assign Store to a Customer</u>	Customer
<u>/api/customers/update/{id}/phone</u>	PUT	<u>Update phone number of a Customer</u>	Customer
<u>/api/staff/post</u>	POST	<u>Add new Staff object to DB</u>	String: Record Created Successfully
<u>/api/staff/lastname/{ln}</u> —	GET	<u>Search Staff by last name</u>	Collection of Staff
<u>/api/staff/firstname/{fn}</u> —	GET	<u>Search Staff by first name</u>	Collection of Staff
<u>/api/staff/email/{email}</u> —	GET	<u>Search Staff by email</u>	Staff
<u>/api/Staff/{id}/address</u> —	PUT	<u>Assign Address to a Staff</u>	Staff & Address details

 Path URL	 HTTP Methods	 Description	 Success Response
<u>/api/staff/city/{city}</u> . —	GET	<u>Search Staff by City</u>	Collection of Staff with Address
<u>/api/staff/country/{country}</u> .	GET	<u>Search Customers by Country</u>	Collection of Staff with Address
<u>/api/staff/phone/{phone}</u> . —	GET	<u>Search Staff by phone number</u>	Staff with Address
<u>/api/staff/update/fn/{id}</u> . —	PUT	<u>Update first name of Staff</u>	Staff
<u>/api/staff/update/ln/{id}</u> . —	PUT	<u>Update last name of Staff</u>	Staff
<u>/api/staff/update/email/{id}</u> . —	PUT	<u>Update email of Staff</u>	Staff
<u>/api/staff/update/store/{id}</u> . —	PUT	<u>Assign Store to a Staff</u>	Staff
<u>/api/staff/update/phone/{id}</u> . —	PUT	<u>Update phone number of a Staff</u>	Staff
<u>/api/store/post</u> . —	POST	<u>Add new Store object to DB</u>	String: Record Created Successfully
<u>/api/store/{storeId}/address/{addressId}</u> .	PUT	<u>Assign Address to a Store</u>	Store & Address details
<u>/api/store/city/{city}</u> . —	GET	<u>Search Store by City</u>	Collection of Store with Address

 Path URL	 HTTP Methods	 Description	 Success Response
<u>/api/store/country/{country}.</u>	GET	<u>Search Store by Country.</u>	Collection of Store with Address
<u>/api/store/phone/{phone}.</u> —	GET	<u>Search Store by phone number</u>	Store with Address
<u>/api/store/update/{storeId}/{phone}.</u>	PUT	<u>Update phone number of a Store</u>	Store
<u>/api/store/{storeId}/manager/{manager_staff_id}.</u>	PUT	<u>Assign manager to a Store</u>	Store & Address details
<u>/api/store/staff/{storeId}.</u> —	GET	<u>Display all Staff of a Store</u>	Collection of Staff
<u>/api/store/customer/{storeId}.</u> —	GET	<u>Display all Customers of a Store</u>	Collection of Customers
<u>/api/store/manager/{storeId}.</u> —	GET	<u>Display Manager Details of a Store</u>	Staff
<u>/api/store/managers.</u> —	GET	<u>Display Manager details (first name, last name, email, phone) and Store details (Address, City, Phone) of all stores</u>	Custom Object
<u>/api/inventory/add</u> —	POST	<u>Add one more Film to a Store</u>	String: Record Created Successfully
<u>/api/inventory/films</u> —	GET	<u>Display inventory(count) of all Films in all Stores</u>	Collection of Film title, no. of copies
<u>/api/inventory/store/{id}.</u>	GET	<u>Display inventory of all Films by a Store</u>	Collection of Film title, no. of copies

 Path URL	 HTTP Methods	 Description	 Success Response
—			
<u>/api/inventory/film/{id}</u>	GET	<u>Display inventory(count) of a Film in all Stores</u>	Collection of Store address, no. of copies
—			
<u>/api/inventory/film/{id}/store/{id}</u>	GET	<u>Display inventory(count) of a Film in a Store</u>	Store Address, no. of copies
—			
<u>/api/rental/add</u>	POST	<u>Rent a Film</u>	String: Record Created Successfully
—			
<u>/api/rental/customer/{id}</u>	GET	<u>Display all Films rented to a customer</u>	Collection of Films
—			
<u>/api/rental/toptenfilms</u>	GET	<u>Display top 10 most rented Films</u>	Collection of Films
—			
<u>/api/rental/toptenfilms/store/{id}</u>	GET	<u>Display top 10 most rented Films of a Store</u>	Collection of Films
—			
<u>/api/rental/duelstore/{id}</u>	GET	<u>Display list of Customers who have not yet returned the Film</u>	Collection of Customers
—			
<u>/api/rental/update/returndate/{id}</u>	POST	<u>Update return date</u>	Rental
—			
<u>/api/payment/add</u>	PUT	<u>Make one Payment entry</u>	String: Record Created Successfully

 Path URL	 HTTP Methods	 Description	 Success Response
<u>/api/payment/revenue/datewise</u> –	GET	<u>Calculate Cumulative revenue of all Stores</u>	Collection of payment_date, amount
<u>/api/payment/revenue/datewise/store/{id}</u> .	GET	<u>Calculate Cumulative revenue of a Store</u>	Collection of payment_date, amount
<u>/api/payment/revenue/filmwise/</u> –	GET	<u>Calculate Cumulative revenue of all Films (across all stores).</u>	Collection of Film title, amount
<u>/api/payment/revenue/film/{id}</u> –	GET	<u>Calculate Cumulative revenue of a Film Store wise</u>	Collection of Store address, amount
<u>/api/payment/revenue/films/store/{id}</u> –	GET	<u>Calculate Cumulative revenue of all Films by Store</u>	Collection Film title, amount